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Checking Counterfeit Critiques on Commodities using Ensemble Classifiers Enhancing Information Credibility

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Abstract

The conundrum of the ubiquitous deceptive reviews has overruled the online ontology with the obsession of obscure but obligatory posting of product reviews for the customers to believe, behold and beget the online product marketing. This mandates contemporary research in the direction to delve deeper on the application and analysis of deceiving online reviews with matured and advanced AI models functional on large scale datasets to effectively and efficiently demarcate between the genuine and the sham. The research counteracts the counterfeiting product reviews via the applications, assessment and analysis of the befitting AI models – Elastic-net Classifier model based on block coordinate descent with Wordcloud and its further performance enhancement through LightGBM Trees Classifier with Grid Search and Early Stopping support, with Log-Loss as performance metric for experimentation to gain insight into the intricacies of detection, diagnosis and diminution of fake product reviews. The paper also delineates discriminative and affirmative aspects of the dataset quality, statistics, stability and standards inherent and coherent to the creation of the dataset using Large Language Models (LLMs) intrinsic to the zeitgeist juncture of recent times promoting machines to produce large scale, cost effective bogus reviews in lieu of the Amazon Mechanical Turks. The results obtained with the Log-Loss holdout score of 0.1462 conforming the LightGBM classifier proves its performance better than the Elastic-Net classifier, conforming it as better than the ROC-AUC in terms of its proximity to the prediction probability for the matching actual/true value.

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1. Introduction

The ubiquitous forged reviews are gaining supremacy as prevalence influencers to online customers, on the ever-demanding online shopping portals. This miraculous growth in online shopping has expedited postage of phony reviews on online products to promote its marketing, purchase and promotion meticulously to the extent that discriminating genuine and bogus reviews have become quite impossible for human intelligence.

This "forged phenomenon" is driven by the contemporary technology serving as basis for fake reviews generation, identification, detection and diminution. The dogmatic dissemination of fake reviews is not only driven by human generated deceptive reviews (like Amazon Mechanical Turks) but also corroborated by computer generated dishonest reviews triggered by incentivized technological advancement in text generation via Natural Language Processing (NLP) and Machine Learning (ML) implicating generative language models like GPT-2 capable of producing fake reviews at a large scale with least cost as compared to humans.

2. Information Credibility - Consequences

The minute, crucial, cognitive and correlated assessment of product reviews using contemporary approaches is a challenge as well as an opportunity to assess information credibility owing to the following reasons:

- Reviews, both positive and negative may be genuine and/or fake. However, intrinsically it serves as a factor of trustworthiness for the customer for their reliance on the brand and the product, not only in the context of recent purchases made but also in terms of future purchases and association with the brand. While positive reviews promote the brand reliability and its product marketing, negative reviews (though may be genuine) impact adversely. Furthermore, the degrading fake reviews likely would breach the customers' trust on the brand and the product, staking the company's reputation resulting in heavy loss in marketing of its products and the business.
- Feedbacks also influence the product rankings on the portal which serve as an important decisive factor for the customers to buy or reject the product online. Moreover, diverse plethora of products and brands existing in each product category exploit the customers for plentiful choices wherein product ranking holds crucial aspect for the customers to proceed with the order. The ranking algorithms are highly influenced by the product reviews and hence, counterfeit reviews are threatening for the brand's survival. From this perspective, deceptive reviews have become highly instrumental for many brands in attacking a brand's credibility, to survive the tough market competition.

3. The Relevance of the Literature Review

3.1. The Conceptual Aspects

The deceptive review implicates illusionary writing about the product without any genuine experience of the same. Moreover, the recent advancement in NLP have made it possible to generate fake reviews at a large scale which have become indistinguishable from human fake inscriptions. "Review spams" are typically attributed for its posting on e-commerce sites and online social media portals for product marketing promotion and staking competitive brands. "Incentivized reviews" are also on rise which are pitched via brand endorsements. Though review spams may be written by an anonymous entity, incentivized reviews are usually written by paid/endorsed opinion spammers who can be traced. Hence, besides opinion spams, the recent research is also driven towards behavioral aspects of opinion spammers and group opinion spammers as a collective effort for fake reviews detection [1] [2].

3.2. The Directions of Sham Analysis

The diagnosis of fake reviews started initially with manual analysis based on heuristic rules [3] incorporating the length, sentiment and helpfulness score to discriminate incentivized vs. non incentivized reviews. Another

investigation [4] focused on the style of writing along with the content of the review implicating the length of review, the degree of detail, presence of pictures and extreme negativity or positivity as factors for fake review detection method. However, fake reviews inspection on heuristic rules are limited by the probability of its accuracy [5] as in case of a single spammer posting reviews occasionally have high likelihood of bypassing the heuristic-based detection techniques. Besides this, heuristic rules aren't subjected to update with the frequency with which the opinion spammers adapt their review content and writing style circumventing the heuristic-based approaches.

The practical implementations of deploying humans as examiners of fake reviews have proved futile based on the studies in [6] [7] [8] where in the human accuracy have been reported to be 65%, 57% and 52% respectively, which has been far lower than the performance reported by ML models (86%) [8].

The massively growing online market has witnessed the scale of products and their reviews increasing exponentially which lie beyond the capacity of manual approaches of fake reviews detection by humans [9]. This drawback manifests the need and necessity of automation techniques for deceptive reviews detection implied and applied by the artificial intelligence approaches to detect and derive patterns from voluminous data samples, for intriguing results, which lies beyond human capacity. The in-depth review of the literature on automated techniques to detect dummy reviews [10] has been categorized into (a) lexicon and linguistic driven content matter in the review text incorporating bag-of-words and n-gram features, POS (Parts Of Speech) and LIWC (Linguistic Inquiry and Word Count), and other stylistic and semantical feature based approaches (b) behavioral characteristics of the opinion spammers like rating eccentricity, review burst detection, content likeness, exciting ratings, postage of reviews in initial time frame etc.(c) relational features like focusing on connotation between the reviewers and products in the context with the review sentiment, retrieving rapport aspects among reviews, reviewers and IP addresses etc. and (d) collusive opinion spammers engaged into group spamming driven towards inspection on spam group cohesion, the count of products as goal of the spammer group, ratio of product and reviewer in the group, magnitude of the spammer group, reviewer correlation in the spammer group etc. [1] [2]

As evident the hybrid approaches promoting combination of features like textual and behavioral, and further its union with feature association of spammer groups are always a better approach to detecting the review spams. Delving into the other hybrid approach, combination of human and machine efforts have also been reported in the literature though such initiatives have been taken by researchers quite rare. [11] accentuates human detection of bogus reviews on the basis of union of textual and contextual aspects of the reviews. [12] developed an amalgamated approach by mixing algorithmically hauled psycho-linguistic features with the decisions reverted by the two machine learning algorithms, that were subjected to the consensus of the human detectors involved in the experiment. This hybrid approach proved significant in improving the machine performance by marginally 0.2 percent establishing the noteworthiness of the human effort vis a vis machine approach towards fake reviews detection as statistically significant [13]

This ordeal connotes the fact that AI based approaches for dummy reviews detection is impossible without the human efforts being extended to train the AI algorithms with the inherent and coherent bias characterized by the human choice of dataset, its preprocessing, choice of classification models, application of feature engineering, and the multiple options of the models hyperparameter tunings to obtain the best possible results [14].

3.3. The Dataset's Contextual Diversity

The availability of datasets on fake reviews is very scarce, each dataset posing a different challenge in the framework of its content, context and application to the AI models for the purpose of classification. [15] extended substantial effort in inspecting Amazon's 10 million online reviews with the objective of detecting the diverse opinion spams. Though the accuracy obtained was 63% on textual and 78% on unification of all the available features, yet the credibility of the results was demarked by accounting replicated and near replicated reviews as fake reviews, which otherwise may have been genuine because of many legitimate reasons. This connotes the fact that many anomalous activities may not be attributed as deceitful happenings. It is hereby noteworthy to mention that the dataset used for the experiment doesn't entail imitated or near imitated reviews, rather it implicates genuine fake reviews which are computer generated and didn't exist in the dataset prior to its generation.

Yet another paper by [16] demystifying the experiential dataset incorporating balanced positive and negative review count as 42, focuses on the reviews' comparative psycho-linguistic differences but, fails to achieve the purpose of training ML classifiers to achieve promising results owing to the very small size of the dataset.

3.4. The Dataset's Deployment Delineation

The publicly available selected dataset [17] has been initially developed by the researchers [13] scrutinizing the aspect of stratified sampling to ascertain arbitrary selection of equal proportion of sample reviews per product class. Top 10 amazon product types with loftiest product reviews (viz. Kindle Store; Pet Inventories; Tools and Home Upgrading; Toys and Games, Cinemas and Television; Sports and Alfresco, Books; Dresses, Shoes, and Jewelry; Home and Kitchen; Electronics) accounting to 88.4% of the appraisals apropos to the precedent dataset, with the intent to reasonably signify the baseline dataset attributed to the Amazon (2018) product reviews dataset [18] have been incorporated in this dataset [13].

Implementing the GPT-2 language model, the researchers [13] have generated 2000 reviews per product category by fine tuning the language model. The model takes into consideration 'w' primary words (by setting w=5, a typical parameter value contributing to text generation tasks [19]) with respect to each tested review and produces rest of the feedbacks automatically. Based on an analysis of the review length distribution (e.g., the number of words contained in reviews in the entire dataset), the researchers have been generating and targeting individual segments with single-word intervals spanning between 10 to 350 words as sentence length of the generated review, as depicted in Fig. 1 [13], superintending the segments as per the original dispersal in the sample. To exemplify, if the proportion of 50-word assessments in the Amazon dataset forms 0.5% of the total reviews then 0.5% of the engendered samples too contain reviews which are 50-words long.

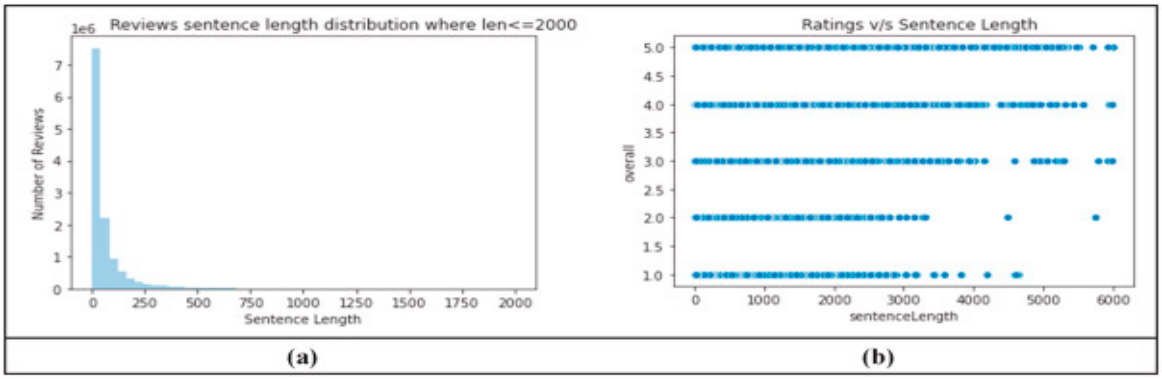


Fig. 1. Review length and rating dialectics of the sample dataset [13]

Fig. 1. (a) depicts sentence length dispersal (M = 305.6 characters, SD = 307.0). The spreading is uneven and the count of appraisals decreases significantly after 350 words. Actually, only about 2.5% of reviews are longer than 2000 words; Fig. 1. (b) As can be seen in the scatterplot, review ratings are not correlated with sentence length. Each score level (1-5) has appraisals of varying extents which doesn't mandate to stratify the sample by class. Explicating in other words the sample dataset complies with two conditions viz. (i) the engendered reviews are approximately similarly segregated over the most prevalent product categories and (ii) it caters to varied length of reviews with similar count of reviews under dissimilar rating levels, to account for both encouraging and discouraging review natures.

As imbalanced datasets significantly negate ML classifiers performance, the selected dataset ensures data balance by populating the dataset with 20,000 artificially generated deceitful reviews presented as computer-generated reviews (CG) and 20,000 reviews connoting to the original reviews (OR) written by humans, retrieved from the Amazon dataset [18], manifesting it as a binary classification problem.

Delving deeper into the selected sampled dataset, Fig. 2. (a) below indicates that the extent of reviews complies with the dispersal of the reviews in the original dataset. In Fig. 2. (b) the two classes signifying computer generated (CG) fake reviews and original reviews (OR) posted by humans taken from the Amazon dataset are equally balanced, nullifying the need of under or oversampling of data for better training of the ML classifiers. In Fig. 2. (c) the review count by product categories is roughly equal. These factors contribute to the learning of ML classifiers withstanding a good probability of class demarcation across varied product classes, in turn adding to the generalizability of the AI model.

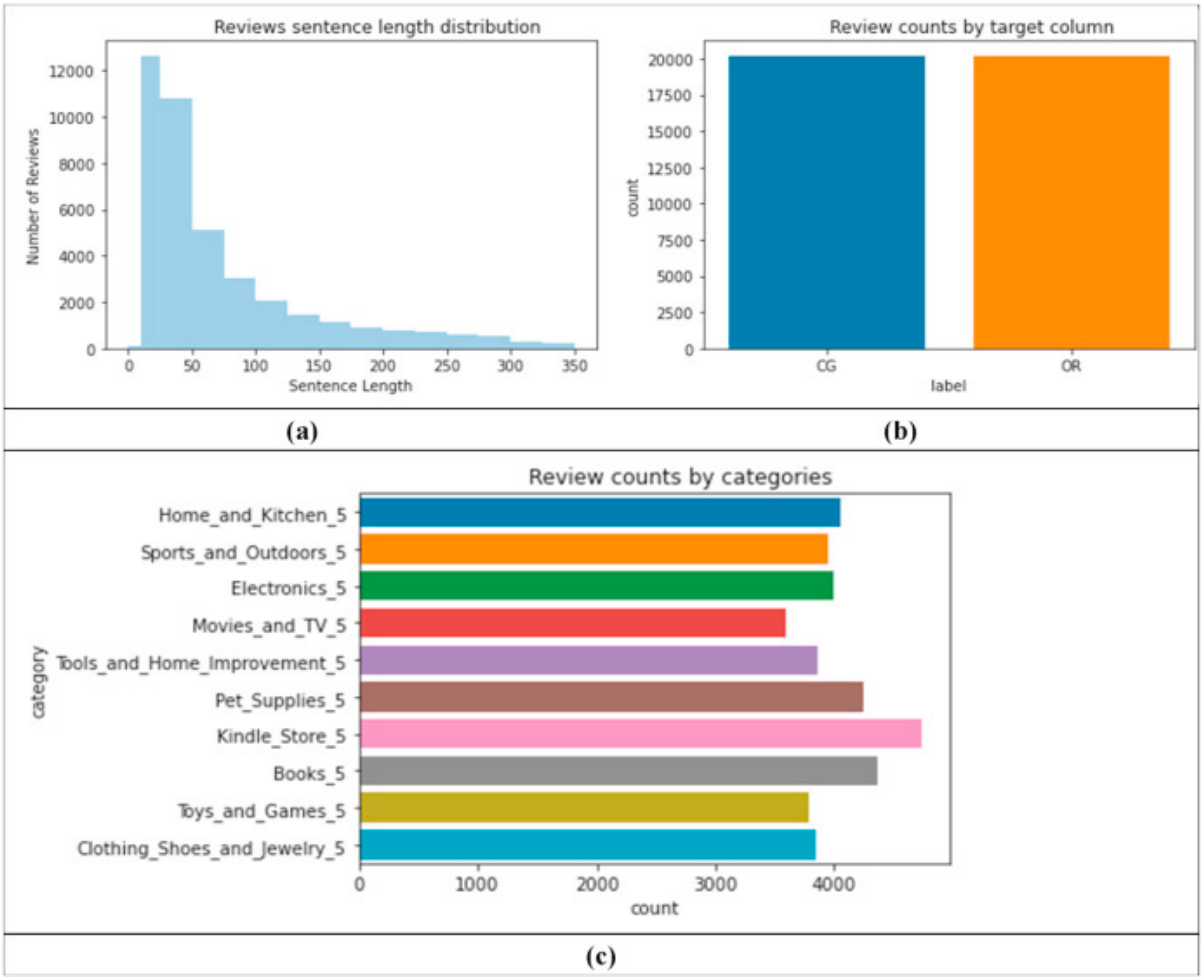


Fig. 2. Sample dataset properties [13]

4. The Experimental Exposition – AI Models Viabilities

This section depicts the proposed methodology as indicated in Fig. 3 below, which illustrates the stages of the experimentation and its emulations under the purview of deployment of different ML classifiers to its advancement into ensemble learning approach in the light of the performance analysis driven by the Log-Loss [20] as evaluation metric. For all the experiments adjudging the models' performances, the commonality exists in terms of the dataset partitioning that follows "stratified sampling" that entails dividing the population into homogenous but disjoint subgroups, known as "strata", with further application of simple random sampling within each subgroup, ensuring the test set being the representative of the population with percentage of each stratum preserved.

The validation type followed is "training-validation-holdout" wherein 64% of the dataset is being used for training the model, 16% of the dataset is being used for validation (that connotes to cross-validation to assess the training quality of the model as the training proceeds and proves highly effective to overcome over-fitting that occurs when the training algorithm tends to focus more on training set details at the cost of losing generalization ability of the model), and 20% of the dataset is being used for testing the final trained model as testing set (or holdout dataset) contributory in assessing the model's performance, as illustrated in Fig. 4 below.

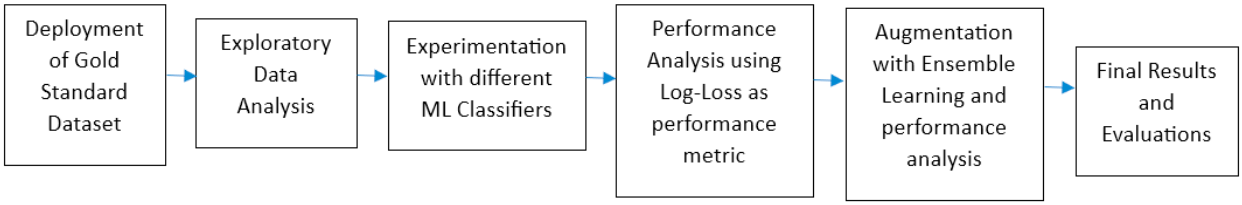


Fig. 3. Proposed Methodology

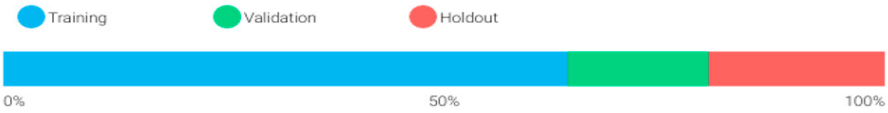


Fig. 4. Dataset demarcation between Training, Validation and Holdout sets

The experiment performed implies to a legitimate set of top two models ranked as per their performance in the binary classification problem of classifying the product reviews as genuine or fake, elucidated in terms of the model's blue print as depicted in the figures viz. Fig. 5 and Fig. 6 below.

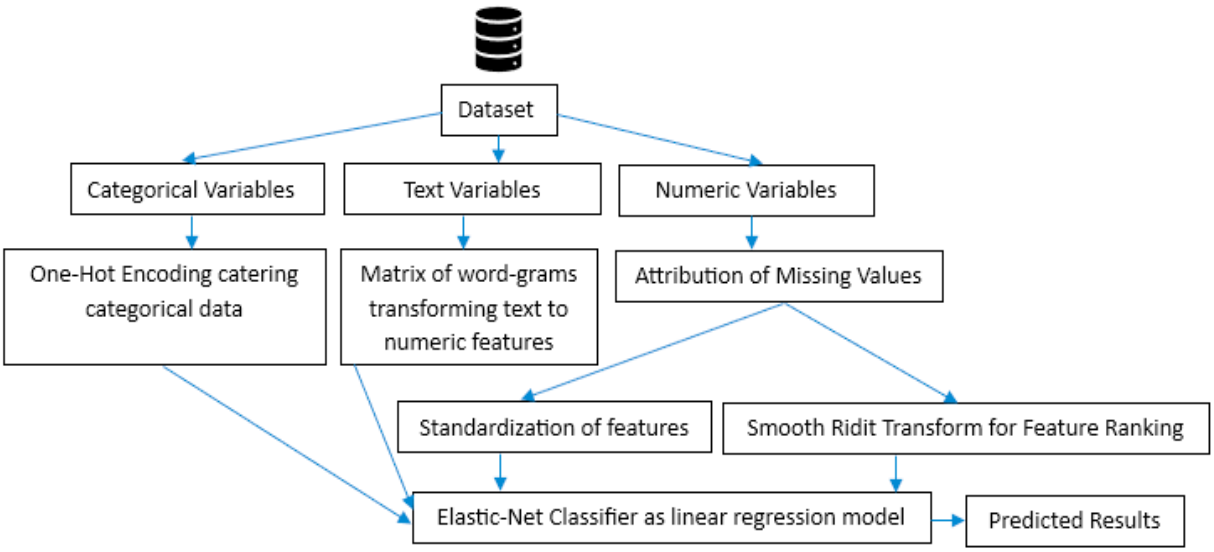


Fig. 5. Elastic-Net Classifier (L2 / Binomial Deviance)

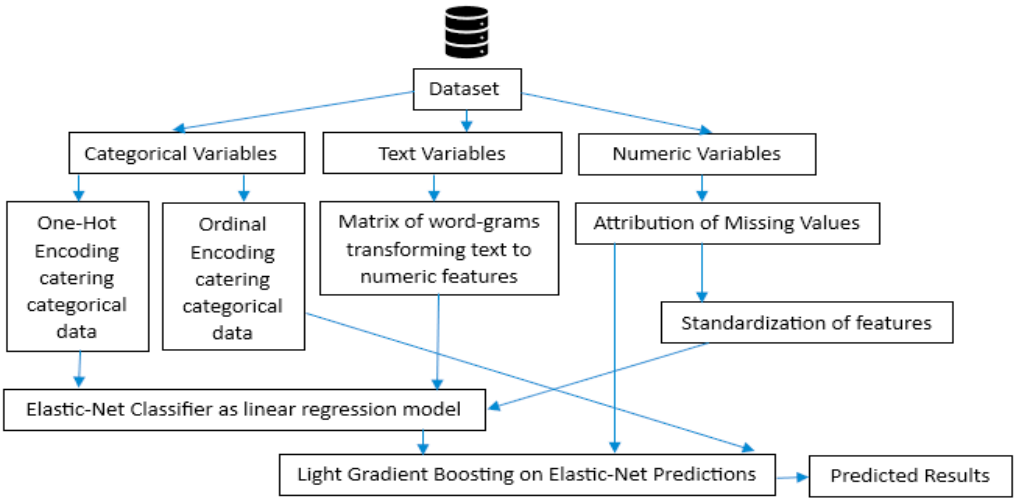


Fig. 6. Light Gradient Boosting on Elastic-Net Predictions

The delimitation of both the above model elements are as follows:

- One-Hot Encoding: Fulfils the need for feeding the categorical data to the ML classifier distinctively that is compatible to the learning of the ML classifier particularly in case of presence of multi-categorical feature in the dataset.
- Missing Values Ascribed: Refers to computation of median feature value via a speedy median procedure (grounded on np.partition).
- Standardize: Standardizes features by eliminating the median and grading to unit standard deviation or the mean absolute deviation.
- Matrix of word-grams occurrences: Refers to method for creating numeric features from a text column.
- Ordinal encoding of categorical variables: Effective for a categorical feature, for converting categorical levels to an ordinal scale.
- Elastic-Net Classifier (L2 / Binomial Deviance): Elastic-Net is a linear regression model proficient with L1 and L2 prior as regularizer [2]. Elastic-Net is useful with multi-correlated features. Lasso randomly picks one of these whereas elastic-net possibly picks both.
- Light Gradient Boosting on Elastic-Net Predictions: This entails LightGBM Trees Classifier with Grid Search and Early Stopping support [21] that implicates a tree-based algorithm and is premeditated to be dispensed and structured, lending itself to the following advantages:
    - Gradient Boosting Machines: They are attributed as an innovative algorithm suitable for tremendously precise predictive models.
    - Early Stopping Support: Connotes to a technique for ascertaining how many trees is to be used for a boosted trees model. The model scores on the test set for each loop unlike the training data, post bifurcation of the data into training and test sets. This serves to be congenial to control overfitting by stopping at the juncture beyond which the trees produced further would not lead to a better accuracy, thereby saving the time for algorithm processing.
    - Grid Search Support: It is attributed to run during training to guesstimate the ideal model parameter values that yield the finest performance (evaluated by the constituted loss function).
- Smooth Ridit Transform: Helps to rank and normalize the numeric features by replacing the numeric values with their relative ranks within each feature impacting in reduction of outliers and uneven distributions.

The above explanation is brief and specific to the role of these elements in models' architecture from a very high-level perspective. Researchers may further delve deeper into the underpinnings to explore the underlying details in an articulated manner [21] [22].

5. The Result Inspection – Experimentation's Revelations

The aftermath of the models experimented above necessitates inclusion of Log-Loss as performance metric [23]. The Log-Loss (a.k.a. cross-entropy loss) is indicative of the proximity of the prediction probability to the corresponding actual/true value, which votes it for being a better performance metric than ROC-AUC [24] as manifested in this research. The lower the log-loss value, the better the model performance. However, both the performance metrics viz. Log-Loss and ROC-AUC evaluation have been presented below to illustrate the models' performance.

Table 1 and Table 2 depict the Log-Loss Performance metric of LightGBM vis a vis Elastic-Net Classifier (L2 / Binomial Deviance) proving the credibility of LightGBM. Table 3 further elucidates the performance of other alternative models that were the candidates of the experimentation process. Table 4 indicates the dataset features taken into consideration by the AI models for the experimentation purposes. Table 5 highlights the other key sensitivity values ranked from top to bottom as per the AI model performance for clarity, completeness and consistency, assisting the comparative analysis of the predictive results.

Table 1. Performance of the Light Gradient Boosting on Elastic-Net Predictions on Log-Loss Performance Metric

Scoring Type	Score (Log Loss)
holdout	0.1462
validation	0.1501

Table 2. Performance of the Elastic-Net Classifier (L2 / Binomial Deviance) Predictions on Log-Loss Performance Metric

Scoring Type	Score (Log Loss)
holdout	0.1463
validation	0.1502

Table 3. Performance of Alternative Models' Predictions on Log-Loss Performance Metric sorted by Holdout score

Model Name	Validation Score	Holdout Score	Training Dataset %
Generalized Additive2 Model	0.1503	0.1462	63.9988
Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)	0.1504	0.1465	63.9988
Light Gradient Boosted Trees Classifier with Early Stopping	0.1517	0.1486	63.9988
eXtreme Gradient Boosted Trees Classifier with Early Stopping	0.1524	0.1488	63.9988
Keras Slim Residual Neural Network Classifier using Training Schedule (1 Layer: 64 Units)	0.2405	0.2477	63.9988

Table 4. Model Features and Summary Statistics

Feature Name	Var Type	Unique	Missing	Mean	Std Dev	Median	Min	Max
category	Categorical	10	0	N/A	N/A	N/A	N/A	N/A
rating	Numeric	5	0	4.26	1.14	5.0	1.0	5.0
label	Categorical	2	0	N/A	N/A	N/A	N/A	N/A
Text	Text	32333	0	N/A	N/A	N/A	N/A	N/A

Table 5. Key Sensitivity Values (inherent to Confusion Matrix) ranked from top to bottom as per the model performance

Model Name	F1 Score	True Positive Rate	False Positive Rate	True Negative Rate	Positive Predictive Value	Negative Predictive Value	Accuracy	Matthews Correlation Coefficient
Light Gradient Boosting on Elastic-Net	0.9412	0.954	0.0732	0.9268	0.9287	0.9527	0.9404	0.8811
Elastic-Net Classifier (L2 /	0.941	0.9565	0.0764	0.9236	0.926	0.955	0.94	0.8805

Binomial Deviance)								
Generalized Additive2 Model	0.9405	0.9392	0.0579	0.9421	0.9419	0.9393	0.9406	0.8813
Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)	0.9409	0.9451	0.0638	0.9362	0.9367	0.9446	0.9406	0.8813
Light Gradient Boosted Trees Classifier with Early Stopping	0.9402	0.957	0.0786	0.9214	0.9241	0.9554	0.9392	0.8789
eXtreme Gradient Boosted Trees Classifier with Early Stopping	0.94	0.9545	0.0764	0.9236	0.9259	0.953	0.939	0.8785
Keras Slim Residual Neural Network Classifier using Training Schedule (1 Layer: 64 Units)	0.8987	0.9147	0.1209	0.8791	0.8832	0.9115	0.8969	0.7942

In relevance to the ROC-AUC depicted in Fig. 7 and Fig. 8 that indicates the same value for the AUC being (0.9869) for both the models, the Log-Loss values are different viz. holdout score (0.1462) and holdout score (0.1463) respectively. This manifests the fact that Log-Loss is better as a performance metric than ROC-AUC. The AUC being (0.9869) for both the models indicate they are outstanding classifiers in distinguishing by predicting the reviews to be original or fake for the given dataset, when tested on holdout dataset.



Fig. 7. ROC – AUC (0.9869) based on the Holdout score (0.1462) of Light Gradient Boosting on Elastic-Net Predictions



Fig. 8. ROC – AUC (0.9869) based on the Holdout score (0.1463) of Elastic-Net Classifier (L2 / Binomial Deviance)

6. Conclusion and Future Work

The realm of online domain attributed to product reviews subjected to deceptive opinion spamming is dominating, decorative, and disseminating diversely into not only the domain of product reviews but also into service reviews, sentiment semantics, star ratings and plethora of faking facets spanning from opinion spams to opinion spammers and the sighting spread as collusive opinion spammers.

The research into this dogmatic area of deceptive review detection has also advanced quite aggressively from assessments based on linguistic features to behavioral and temporal features involving group spamming activities, complemented with stylistic and stylometric features of the reviews. Gradually the ML models' applications have also matured with ensemble approaches straddling from machine learning to deep learning model implementations.

The research is an endeavor to experiment with advanced AI models comprising combination of methodologies apt to the model integration for producing better promising results. Two of the superlative accomplished models have been explicated with a glimpse into other alternative models evaluated on the performance metric for the

understanding on the nuances of the finest executed models. The dataset has also been experimented with its division into train, validate and test sets unlike just the train and test set which has been a general norm for dataset distribution for the training of ML models, from the perspective to control overfitting.

Future work entails improvement on the models' performance based on hyperparameter tuning to enhance the results further. This may change the ranking of the models in terms of their performance. Performance metrics combination and comparisons are yet other factors to explore, experiment and enhance.

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